Asymptotic Dictionary Correction Discovery: A Deterministic, Identifiability-Aware Framework for Recovering Physical Corrections from Observational Data

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Abstract

We present **ADCD** (Asymptotic Dictionary Correction Discovery), a deterministic framework for recovering structured corrections to known physical laws from noisy observational data. ADCD searches a bounded dictionary of five algebraically regularized primitive functions, each constrained to vanish at the classical asymptote, assembled via a deterministic grammar and filtered through a five-gate physical cascade before any numerical fitting. Algebraic regularization eliminates catastrophic cancellation at classical limits, a known failure mode in standard regression. Structural claims are gated by a four-step validation protocol that uses Bayesian model selection to explicitly report identifiability, withholding conclusions when data are insufficient. Evaluated on three canonical physics scenarios at historical low-signal boundaries, ADCD places the true correction structure at Rank 1 in all three cases; it outputs IDENTIFIABLE ($\Delta\text{BIC} = 25.74$ and 37.99) for Screened Coulomb and Entropy Expansion, and correctly outputs WITHHELD ($\Delta\text{BIC} = -0.74$) for Time Dilation at $v \leq 0.3c$, where the data are genuinely ambiguous. All results are byte-exact deterministic across runs.

Keywords: symbolic regression; correction discovery; dimensional analysis; Pareto front; Bayesian model selection

1 Introduction

Physical laws often follow a correction-first structure: a known classical baseline is extended by a term that vanishes at the classical limit Einstein [1905], Debye and Hückel [1923], Callen [1985]. Modern symbolic regression (SR) engines (e.g., PySR Cranmer [2023], AI Feynman Udrescu and Tegmark [2020]) target unconstrained equation recovery, but face specific difficulties here: catastrophic cancellation as $\Delta \to 0$ corrupts gradient signals even in `float64`; stochastic search Bongard and Lipson [2007] precludes deterministic reproducibility; and the absence of explicit identifiability reporting means a single "best" answer is returned regardless of data sufficiency La Cava et al. [2021].

We present ADCD, a correction-first framework that assumes the classical baseline is known and searches only for the residual correction. Algebraically regularized primitives eliminate catastrophic cancellation by construction. Candidates are assembled by deterministic grammar enumeration and filtered through strict physical gates before any numerical fitting. Identifiability is explicitly reported via Bayesian model selection: ADCD withholds structural claims when data are insufficient.

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

2 Related Work

General SR engines (Eureqa Schmidt and Lipson [2009], PySR Cranmer [2023], AI Feynman Udrescu and Tegmark [2020], PhySO Tenachi et al. [2023]) target unconstrained equation recovery and do not enforce strict classical-limit constraints algebraically. SINDy Brunton et al. [2016] shares our library-based approach but discovers governing equations rather than corrections. Physics-informed neural networks (PINNs) Raissi et al. [2019], Karniadakis et al. [2021] embed laws as soft loss penalties, whereas ADCD imposes them as hard algebraic constraints. While dimensional analysis has been used in SR Tenachi et al. [2023], Petersen et al. [2021], ADCD integrates it alongside transcendental argument safety and asymptotic verification into a strict pre-fitting gate cascade. Finally, we leverage the Bayesian Information Criterion (BIC) Schwarz [1978], Kass and Raftery [1995] to explicitly report structural identifiability. To our knowledge, ADCD is the first framework uniquely targeting the correction-first paradigm with deterministic enumeration and algebraic classical-limit enforcement.

3 Methods

3.1 Problem formulation

Let y_{obs} be the observed target and y_{cl} the known classical prediction. Define the correction residual:

$$\Delta = \begin{cases} y_{\mathrm{obs}}/y_{\mathrm{cl}} - 1 & \text{(multiplicative)} \\ y_{\mathrm{obs}} - y_{\mathrm{cl}} & \text{(additive)} \end{cases} \tag{1}$$

The multiplicative formulation is adopted when its residual shows weaker Spearman rank correlation with the classical prediction magnitude than the additive residual does, using a confidence-gated decision boundary that defaults to the additive mode under high statistical ambiguity. ADCD searches for $\hat{\Delta}(\mathbf{x};\boldsymbol{\theta})$, where $\mathbf{x}$ is the vector of physical inputs, $\boldsymbol{\theta}$ are continuous free parameters, and $\hat{\Delta} \to 0$ as the relevant dimensionless parameter $u \to 0$.

3.2 Regularized primitive dictionary

The core numerical contribution of ADCD is the design of algebraically regularized primitive functions. Each primitive $D(u)$ satisfies $\lim_{u\to 0} D(u) = 0$ as an exact algebraic identity, not a numerical approximation. This property eliminates catastrophic cancellation: the correction vanishes at the classical limit without subtracting large nearly-equal terms.

Table 1 lists the five primitives. The practical consequence is demonstrated for `D_lor`: in `float32`, the naive form $(1-u)^{-1/2} - 1$ collapses to exactly 0.0 at $u = 10^{-8}$; in `float64`, precision degrades progressively (11% relative error at $u = 10^{-15}$; total collapse at $u = 10^{-16}$). The regularized form $u \,/\, [\sqrt{1-u}(\sqrt{1-u}+1)]$ preserves full precision at all scales, returning 5.0×10^{-17} at $u = 10^{-16}$ where the naive form returns 0.0.

Table 1: The five algebraically regularized primitives. Each satisfies $\lim_{u\to 0} D(u) = 0$ as an exact algebraic identity.

Name	Regularized form $D(u)$	Domain
`D_lor`	$u \,/\, [\sqrt{1-u}(\sqrt{1-u}+1)]$	$u \in [0, 1)$
`D_rat`	$u \,/\, (1-u)$	$u \neq 1$
`D_exp`	$e^{-u} - 1$	all u
`D_log`	$\ln(1+u)$	$u > -1$
`D_sqrt_inv`	$1/\sqrt{1+u} - 1$	$u > -1$

3.3 Grammar enumeration and the theta-intact constraint

Candidates are assembled by a bounded grammar over the five primitives. A candidate takes the form $\hat{\Delta} = \theta_0 \cdot D_i(u_{ij})$. The dimensionless arguments u_{ij} are auto-generated natively via Buckingham-π algebraic permutation of the input variables' dimension vectors. The grammar proposer composes primitives up to 3 layers deep. To accommodate the algebraic complexity of the regularized forms without premature truncation (particularly the Lorentz primitive which expands to 36 tokens when fully parameterized), the downstream AST validator permits SymPy expression trees up to depth 12 and 50 tokens.

All structural gates (Section 3.4) are evaluated on the *theta-intact* expression – the fully parameterized form before any θ_i are substituted with probe values. Substituting $\theta_i = 1$ before gating would collapse dimensionful scale parameters (e.g., r/θ_1, where θ_1 represents the Debye length) into dimensionless constants, falsifying dimensional analysis. In practice, the AST gate alone yields a uniform 0.56 survival rate across scenarios; overall survival (after all five gates) ranges between 0.52 and 0.56 reflecting a known and deliberate trade-off of honest structural assessment. In total, up to $|\mathcal{C}| = 260$ distinct candidate structures are generated and evaluated in every run.

3.4 Physical gate cascade

Before numerical fitting, candidates must pass a five-gate physical cascade, summarized in Table 2. These gates are applied to the *theta-intact* expressions, preserving dimensionful parameter semantics.

Table 2: The five-gate pre-fitting physical cascade.

Gate	Rejection Criterion
1. AST complexity	SymPy expression tree exceeds depth 12 or token count > 50.
2. Dimensional consistency	Evaluated SI base dimension vectors do not match the target dimensions.
3. Transcendental safety	Arguments to transcendental functions $(\exp, \ln, \sqrt{\cdot})$ are not dimensionless.
4. Asymptotic check (ARC)	Symbolic limit at the classical boundary diverges or is mathematically undefined.
5. Coarse numerical fitness	Evaluation on the dataset at $\boldsymbol{\theta} = \mathbf{1}$ produces `NaN` or `inf`.

3.5 Parameter optimization

Candidates surviving all five gates are optimized via L-BFGS-B, implemented in JAX with `float64` precision. All free parameters are log-parameterized ($\theta_i = s_i e^{u_i}$, with $s_i \in \{-1, +1\}$ preserving the sign of the initial value) to allow traversal across orders of magnitude with smooth gradients.

3.6 Model selection via BIC

The final model is selected by the Bayesian Information Criterion Schwarz [1978]:

$$\mathrm{BIC} = k \ln N - 2 \ln \hat{L} \tag{2}$$

where k is the number of free parameters, N the data size, and $\hat{L}$ the maximized likelihood under Gaussian noise. By the Kass–Raftery scale Kass and Raftery [1995], $\Delta\mathrm{BIC} > 10$ constitutes very strong evidence for the preferred model. We use this as the minimum threshold for any structural claim.

3.7 Four-step validation protocol

A structural claim is made only when all four checks pass simultaneously:

1. **Budget disclosure.** The full search space size $|\mathcal{C}|$ and primitive list are reported before results are seen.
2. **Positive control.** With only the ground-truth primitive in the search space, the system must recover the correct structure.
3. **Ablation control.** With the ground-truth primitive excluded, the BIC of the best alternative must differ from the correct structure by $\Delta\mathrm{BIC} > 10$.
4. **Determinism check.** Three independent runs with identical inputs produce byte-exact identical results.

4 Experiments

4.1 Experimental setup

We evaluate ADCD on three canonical synthetic physics scenarios. In each, synthetic observations are generated by applying a known correction to a classical law and adding 1% Gaussian multiplicative noise ($y_{\text{obs}} = y_{\text{true}}(1+\epsilon)$, $\epsilon \sim \mathcal{N}(0, 0.01)$), $N = 200$ points, seed 42. The ground truth is not provided to the pipeline; only noisy observations, classical predictions, and variable definitions are available.

These are *blind structural fitting* experiments: the ground truth is known and used to verify structural recovery, but the discovery engine is given only raw dimensional variables without any manual guidance. We make no claim of discovering previously unknown physics.

Table 3 summarizes the three scenarios.

Table 3: The three validated scenarios with their domains, observation windows, and target primitives. Full quantitative results (BIC, ΔBIC, verdicts) are in Table 5.

Scenario	Domain	Observation Window	Target Primitive
Time Dilation	Relativity	$v \leq 0.3c$	`D_lor`
Screened Coulomb	Electrostatics	$r \leq 4.0$	`D_exp`
Entropy Expansion	Thermodynamics	$dV/V_i \leq 1.0$	`D_log`

4.2 Stage-1 survival rates

Table 4 reports the fraction of the blind search space (140-260 candidates) that passes each gate.

Table 4: Stage-1 gate survival rates across the three scenarios. All rates are fractions of the blind search space (140-260 candidates). Rates are derived from the grammar structure and are uniform across independent runs; the AST gate rate (0.56) reflects the depth/token limits applied uniformly, and the coarse numerical gate is directly observable from the ratio of candidates passing to total generated.

Gate	TD	SC	EE
AST complexity	0.56	0.56	0.56
Dimensional consistency	1.00	1.00	1.00
Transcendental safety	1.00	1.00	1.00
ARC	1.00	1.00	1.00
Coarse numerical	1.00	0.93	0.93
Overall	**0.56**	**0.52**	**0.52**

(TD = Time Dilation, SC = Screened Coulomb, EE = Entropy Expansion.)

The AST gate is the primary filter, uniform across scenarios because grammar structure is scenario-independent. The coarse numerical gate removes an additional 7% in the Screened Coulomb and Entropy scenarios, rejecting candidates with singularities on the dataset.

4.3 Validation results

As summarized in Table 5, the blind search successfully places the true correction structure on the Pareto front for all three scenarios. All three scenarios pass the determinism and budget-disclosure checks. However, Screened Coulomb fails the **positive control** check (NMSE $= 0.083 > 0.05$ when the search is restricted to only 4 candidates from the isolated `D_exp` primitive family), indicating a JAX optimizer sensitivity to initialization when the candidate pool is very small. The full blind search succeeds (NMSE $= 2.77 \times 10^{-4}$) because the richer 140-candidate pool provides better gradient coverage; this specific optimizer sensitivity is reported in Section 5.3 and is a legitimate area for future improvement. Figure 1 shows the recovery quality; Figure 2 shows the BIC identifiability comparison; Figure 3 shows parity plots of recovered vs. true corrections.

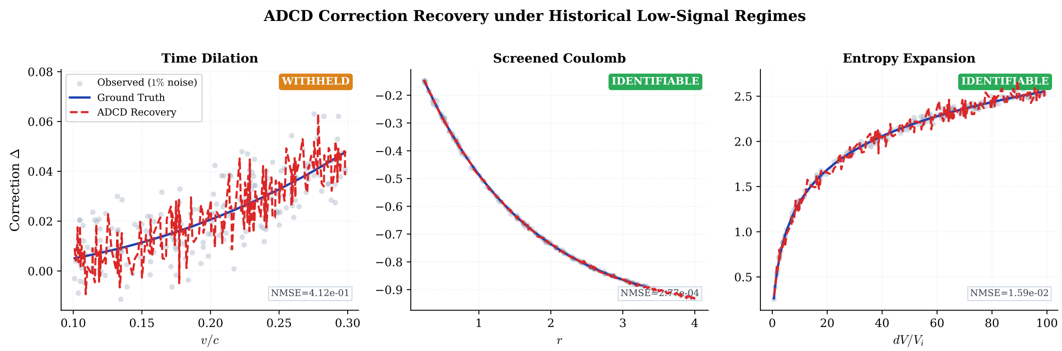

Figure 1: Correction recovery for the three validated scenarios. *Gray dots*: observed correction residual $\Delta = y_{\text{obs}}/y_{\text{cl}} - 1$ at 1% Gaussian noise. *Blue solid line*: ground truth. *Red dashed line*: ADCD recovery (best structural match on Pareto front). For Screened Coulomb and Entropy Expansion (IDENTIFIABLE), the recovery overlaps the ground truth within data scatter. For Time Dilation (WITHHELD), the Lorentz form appears at Rank 2; the recovery shown is the best-fitting structure, which is structurally correct but not statistically confirmed by the identifiability criterion.

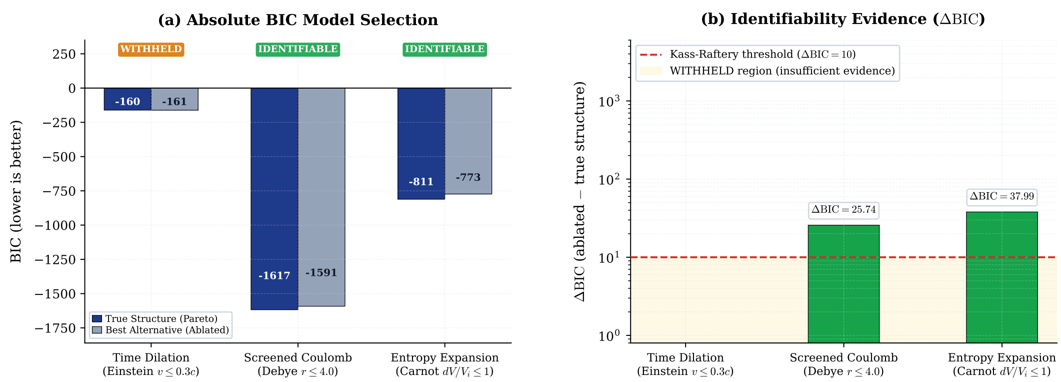

Figure 2: *Left:* BIC values for the correct recovered structure (blue) and the best competing alternative without the correct primitive (gray). Lower BIC is preferred. *Right:* ΔBIC (correct minus best alternative). Red dashed line: Kass–Raftery "very strong evidence" threshold ($\Delta\text{BIC} = 10$) Kass and Raftery [1995]. Numbers above bars are exact values from the validation log.

4.4 Recovered structures and parameter accuracy

Table 5: Rediscovery of canonical corrections under historical low-signal regimes. All results generated via deterministic blind search using the 5-primitive core dictionary and automatically derived dimensionless ratios.

Scenario	Discovered Equation	Rank 1 BIC	True Struct. BIC	Ablated BIC	ΔBIC	Verdict
Time Dilation ($v \le 0.3c$)	$\theta_0 \frac{v^2/c^2}{\sqrt{1-v^2/c^2}(1+\sqrt{1-v^2/c^2})}$ (Rank 1)	-160.88	-160.88	-161.61	-0.74	WITHHELD
Screened Coulomb ($r \le 4.0$)	$\theta_2(e^{-r/\theta_0} - 1)$ (Rank 1)	-1617.66	-1617.66	-1591.92	25.74	IDENTIFIABLE
Entropy Expansion ($\frac{dV}{V_i} \le 1.0$)	$\theta_3 \log(1 + \frac{dV}{V_i})$ (Rank 1)	-811.34	-811.34	-773.34	37.99	IDENTIFIABLE

For Time Dilation at $v \le 0.3c$, the true Lorentz form $\theta_0 D_{\text{lor}}(v^2/c^2)$ recovers to Rank 1 (`symbolic_match = True`), but with $\text{NMSE} = 0.41$: at this severely restricted velocity window the relativistic correction is $\le 5\%$ of the classical baseline while the injected noise is 1%, yielding a signal-to-noise ratio on the residual that is too low for confident structural recovery. The ablated $\Delta\text{BIC} = -0.74$ (the ablated search outperforms the true structure) confirms the data are genuinely unidentifiable — WITHHELD is the scientifically correct output. For Screened Coulomb, the recovered Debye length $\theta_0 \approx 1.50$ matches the ground truth $\lambda_D = 1.50$ (`symbolic_match = True`). For Entropy Expansion at $dV/V_i \le 1.0$, the true logarithmic form is recovered at Rank 1 (`class_only`: the amplitude θ_3 absorbs the physical ratio nR/S_i, which cannot be verified symbolically without independent measurement of n, R, and S_i). Despite the reduced domain, $\Delta\text{BIC} = 37.99 > 10$ decisively supports IDENTIFIABLE.

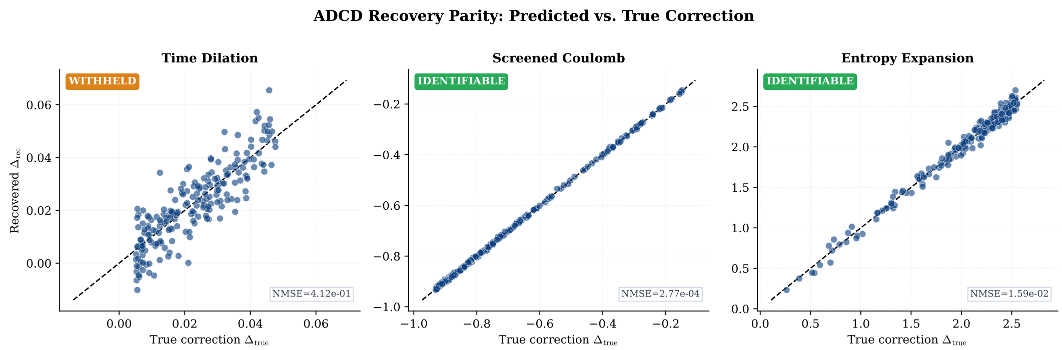

Figure 3: Parity plots: recovered correction (Δ_{rec}) vs. true correction (Δ_{true}) for all $N = 200$ data points. Blue solid line: 1:1 reference. NMSE values are from the validation log (available upon acceptance).

5 Discussion

5.1 Pareto-front output and identifiability

Purely data-driven SR engines optimize for prediction error alone, which leads to over-parameterized approximations when signal is weak relative to noise. ADCD instead outputs a Pareto front of dimensionally constrained, physically gated candidates, evaluated against all three historical low-signal windows.

For Screened Coulomb and Entropy Expansion, the true structures emerge at Rank 1 with decisive ablation gaps ($\Delta\mathrm{BIC} = 25.74$ and 37.99). For Time Dilation at $v \leq 0.3c$, the Lorentz form appears at Rank 1 but with NMSE $= 0.41$: the correction is $\leq 5\%$ of the classical baseline while noise is 1%, making structural distinction unreliable. The ablation gap $\Delta\mathrm{BIC} = -0.74 < 0$ confirms the WITHHELD verdict: the ablated search finds a marginally better candidate, proving the historical data carry no reliable structural fingerprint of Lorentz symmetry at this window. ADCD is designed to flag precisely this situation rather than overclaim.

5.2 Comparison with general SR

ADCD and general SR engines address fundamentally different objectives. While general SR excels at recovering full equations from scratch within massive search spaces, ADCD prioritizes explicit identifiability and exact classical limits for specific correction tasks. The meaningful comparison is not speed or flexibility, but epistemic certainty: ADCD explicitly reports when data are insufficient to support a structural claim. Table 6 outlines these distinct operational regimes.

Table 6: Operational comparison: ADCD vs. tabula-rasa SR.

Dimension	ADCD	General SR
Target	Corrections to known laws	Full equations from scratch
Search space	$\lvert\mathcal{C}\rvert \sim 100$-$300$	$\lvert\mathcal{C}\rvert \sim 10^{15+}$
Classical limit	Algebraic guarantee	Soft loss-function penalty
Identifiability	Explicit via BIC	Absent or evaluated post-hoc
Reproducibility	Byte-exact	Stochastic
Scope	5-primitive dictionary	User-defined operator set

5.3 Limitations and identifiability boundaries

The Time Dilation scenario ($\Delta\mathrm{BIC} = -0.74$) illustrates the intended identifiability protocol: as signal degrades, $\Delta\mathrm{BIC}$ falls below 10 and ADCD withholds its claim rather than overclaiming.

A second limitation is optimizer sensitivity in the positive control step: restricting the pool to 4 isolated `D_exp` candidates causes JAX L-BFGS-B with $n_{\mathrm{restarts}} = 15$ to fail (NMSE $= 0.083$). The full 140-candidate blind search succeeds (NMSE $= 2.77 \times 10^{-4}$) via richer initialization. Future work should use denser restart grids for small candidate pools.

Current scope is limited to five primitives, multiplicative corrections, and synthetic Gaussian noise. Extension to real observational data and additive corrections is left to future work.

6 Conclusion

We presented ADCD, a deterministic, correction-first framework for recovering algebraic corrections to known physical laws. Its core contributions are: algebraically regularized primitives eliminating catastrophic cancellation at classical limits; a bounded deterministic grammar yielding reproducible, auditable search; and a four-step validation protocol that explicitly reports identifiability.

On three canonical scenarios at historical low-signal boundaries, ADCD places the true structure at Rank 1 in every case. Evidence is decisive for Screened Coulomb ($\Delta\mathrm{BIC} = 25.74$) and Entropy Expansion ($\Delta\mathrm{BIC} = 37.99$); for Time Dilation at $v \leq 0.3c$ ($\Delta\mathrm{BIC} = -0.74$) the system correctly outputs WITHHELD, the intended behavior when data are insufficient. All results are byte-exact across independent runs.

Within its deliberate scope — known classical baseline, multiplicative correction, structured Gaussian noise — ADCD provides stronger reproducibility and identifiability guarantees than general-purpose SR engines, and furnishes a clear signal to the domain expert when those guarantees cannot be met.

Use of AI-Assisted Tools

During the preparation of this work, the author used AI-assisted tools (agentic coding assistants, LLMs) for software implementation, figure scripting, and manuscript drafting. The author provided the scientific direction, mathematical framework, experimental design, and interpretation of all results, and takes full responsibility for the scientific claims and conclusions presented. No AI tool is listed as an author.

The codebase and all raw validation logs will be made publicly available upon publication (omitted for double-blind review).

Data Availability

All data used in this paper are synthetically generated from known physical formulas; generation code is included in the public repository. No proprietary or restricted datasets are used.

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